

```
In [132]: import pandas as pd
```

```
In [133]: import mysql.connector
```

```
In [134]: !pip install mysql-connector-python
```

Requirement already satisfied: mysql-connector-python in c:\users\lenovo\anaconda3\lib\site-packages (9.7.0)

```
In [135]: df =pd.read_csv(r"C:\Users\Lenovo\Desktop\finalagridata11
```

```
In [136]: df.head()
```

Out[136]:

	plot_id	location	size	season	crop_type	yield_ar
0	108	West Bengal	1490.00	Kharif	Moong(Green Gram)	
1	113	West Bengal	1279.00	Kharif	Other Kharif pulses	
2	124	West Bengal	1555.00	Kharif	Soyabean	
3	290	West Bengal	2606.00	Kharif	Arhar/Tur	
4	298	West Bengal	9398.00	Kharif	Ginger	

5 rows × 24 columns

```
In [145]: df[["totalcostper(hect)","yield_amount"]].head()
```

Out[145]:

	totalcostper(hect)	yield_amount
0	4750.00	0.55
1	3282.00	0.59
2	3815.00	1.20
3	3307.00	0.76
4	4519.00	1.45

In [146]:

```
df.head()
```

Out[146]:

	plot_id	location	size	season	crop_type	yield_ar
0	108	West Bengal	1490.00	Kharif	Moong(Green Gram)	
1	113	West Bengal	1279.00	Kharif	Other Kharif pulses	
2	124	West Bengal	1555.00	Kharif	Soyabean	
3	290	West Bengal	2606.00	Kharif	Arhar/Tur	
4	298	West Bengal	9398.00	Kharif	Ginger	

5 rows × 24 columns

In [147]:

```
df.head()
```

Out[147]:

	plot_id	location	size	season	crop_type	yield_ar
0	108	West Bengal	1490.00	Kharif	Moong(Green Gram)	
1	113	West Bengal	1279.00	Kharif	Other Kharif pulses	
2	124	West Bengal	1555.00	Kharif	Soyabean	
3	290	West Bengal	2606.00	Kharif	Arhar/Tur	
4	298	West Bengal	9398.00	Kharif	Ginger	

5 rows × 24 columns

```
In [148]: df["yield_amount"].isnull().any()
```

Out[148]: np.False_

```
In [149]: df_encod = pd.get_dummies(df, columns = ["crop_type", "season"])
df_encod.head()
```

Out[149]:

	plot_id	location	size	yield_amount	fertilizer_used
0	108	West Bengal	1490.00	0.55	141803.00
1	113	West Bengal	1279.00	0.59	121722.00
2	124	West Bengal	1555.00	1.20	147989.00
3	290	West Bengal	2606.00	0.76	257473.00
4	298	West Bengal	9398.00	1.45	928522.00

5 rows × 69 columns

In []:

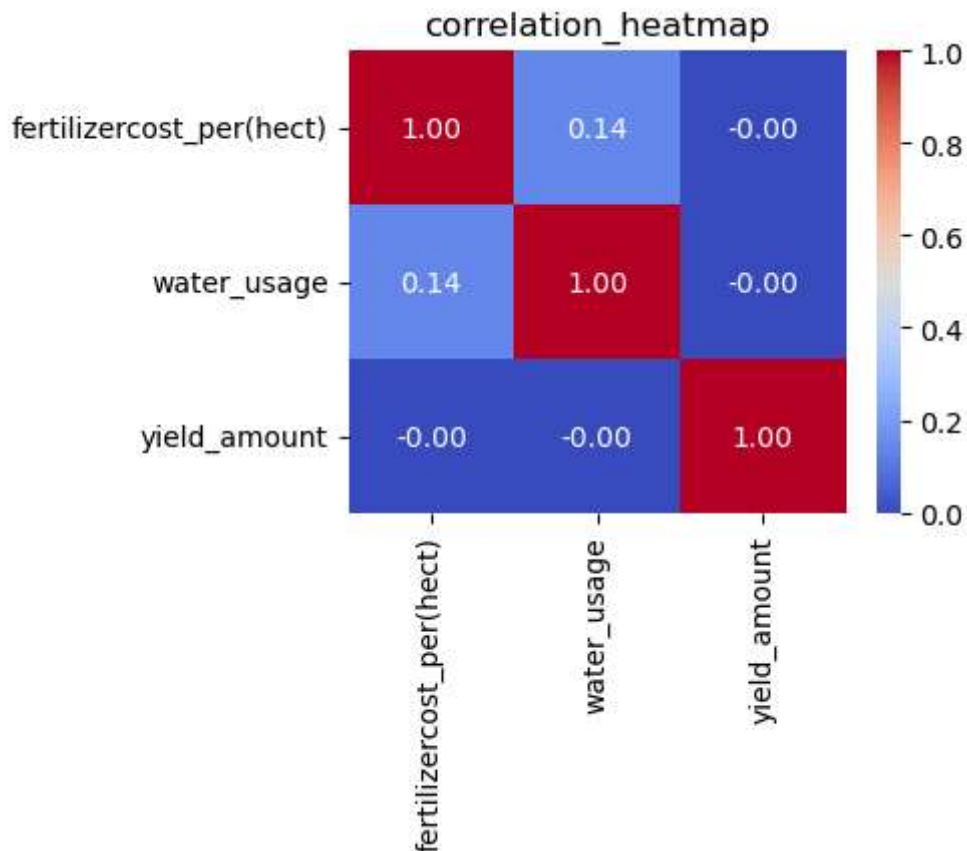
```
In [150]: import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [151]: correlation= df[["fertilizercost_per(hect)","water_usage
```

```
In [152]: corr_matrix = correlation.corr()
print(corr_matrix)
```

		fertilizercost_per(hect)
water_usage	yield_amount	
fertilizercost_per(hect)		1.00
0.14	-0.00	
water_usage		0.14
1.00	-0.00	
yield_amount		-0.00
-0.00	1.00	

```
In [153]: plt.figure(figsize = (4,3))
sns.heatmap(corr_matrix,annot = True,
            cmap = "coolwarm",fmt= ".2f")
plt.title("correlation_heatmap")
plt.savefig("correlation_heatmap.png",
            bbox_inches = "tight")
plt.show()
```



```
In [154]: yield_corr = corr_matrix["yield_amount"].sort_values(ascending=True)
print("\ncorrelation with yield amount:")
print(yield_corr)
```

```
correlation with yield amount:
yield_amount          1.00
water_usage           -0.00
fertilizercost_per(hect) -0.00
Name: yield_amount, dtype: float64
```

```
In [155]: plt.savefig("correlation_heatmap.png",
                    bbox_inches = "tight")
```

<Figure size 640x480 with 0 Axes>

```
In [156]: df.head()
```

Out[156]:

	plot_id	location	size	season	crop_type	yield_ar
0	108	West Bengal	1490.00	Kharif	Moong(Green Gram)	
1	113	West Bengal	1279.00	Kharif	Other Kharif pulses	
2	124	West Bengal	1555.00	Kharif	Soyabean	
3	290	West Bengal	2606.00	Kharif	Arhar/Tur	
4	298	West Bengal	9398.00	Kharif	Ginger	

5 rows × 24 columns

```
In [157]: from sklearn.cluster import KMeans
          from sklearn.preprocessing import StandardScaler
          import matplotlib.pyplot as plt
```

```
In [158]: df_encoded = pd.get_dummies(df, columns = ["crop_type", "season"])
df_encoded.head()
```

Out[158]:

	plot_id	location	size	yield_amount	fertilizer_used
0	108	West Bengal	1490.00	0.55	141803.00
1	113	West Bengal	1279.00	0.59	121722.00
2	124	West Bengal	1555.00	1.20	147989.00
3	290	West Bengal	2606.00	0.76	257473.00
4	298	West Bengal	9398.00	1.45	928522.00

5 rows × 69 columns

```
In [167]: x= df[["totalcostper(hect)", "yield_amount"]]
```

```
In [168]: scaler = StandardScaler()
x_scaled = scaler.fit_transform(x)
```

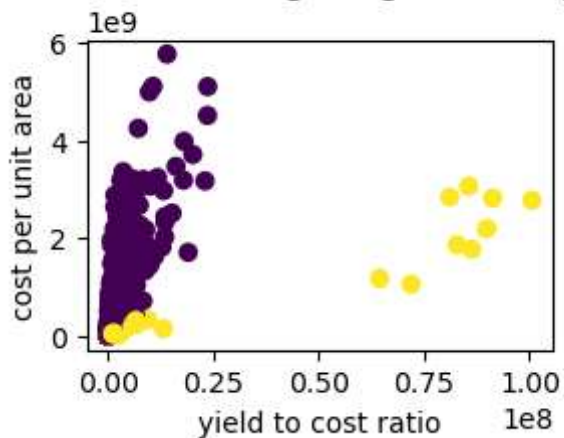
```
In [169]: kmeans= KMeans(n_clusters= 2, random_state = 42)
df["cluster"]= kmeans.fit_predict(x_scaled)
```

```
In [170]: print(df.groupby("cluster")[["totalcostper(hect)", "yield_"])
```

	totalcostper(hect)	yield_amount
cluster		
0	4262.40	1.18
1	3957.09	14.75

```
In [171]: plt.figure(figsize =(3,2))
plt.scatter(
    df["STATEYIELD(QUI)"],
    df["totalcost(state)"],
    c=df["cluster"])
plt.xlabel("yield to cost ratio")
plt.ylabel("cost per unit area")
plt.title("kmeans clustering of agricultural plots")
plt.show()
```

kmeans clustering of agricultural plots



```
In [172]: centroids = pd.DataFrame(
    kmeans.cluster_centers_,
    columns= ["yield_to_cost_scaled", "cost_per_unit_scaled"])
print(centroids)
```

	yield_to_cost_scaled	cost_per_unit_scaled
0	0.02	-0.18
1	-0.37	3.13

```
In [173]: df.head()
```


Out[173]:

	plot_id	location	size	season	crop_type	yield_ar
0	108	West Bengal	1490.00	Kharif	Moong(Green Gram)	
1	113	West Bengal	1279.00	Kharif	Other Kharif pulses	
2	124	West Bengal	1555.00	Kharif	Soyabean	
3	290	West Bengal	2606.00	Kharif	Arhar/Tur	
4	298	West Bengal	9398.00	Kharif	Ginger	

5 rows × 25 columns

```
In [175]: print(df["totalcostper(hect)"].describe())
```

```
count    1000.00
mean     4246.22
std       786.87
min      2528.00
25%      3635.00
50%      4264.00
75%      4853.75
max      5956.00
Name: totalcostper(hect), dtype: float64
```

```
In [176]: df.head()
```

Out[176]:

	plot_id	location	size	season	crop_type	yield_ar
0	108	West Bengal	1490.00	Kharif	Moong(Green Gram)	
1	113	West Bengal	1279.00	Kharif	Other Kharif pulses	
2	124	West Bengal	1555.00	Kharif	Soyabean	
3	290	West Bengal	2606.00	Kharif	Arhar/Tur	
4	298	West Bengal	9398.00	Kharif	Ginger	

5 rows × 25 columns

In []: